

Math and 3D Obligatory 1 Submission

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Abstract

Full disclosure : my code did not compile properly all the way until February 3rd, and it's a miracle that anything is submitted at this point.

1 Introduction

In this compulsory we were tasked with solving equations given to us at Math and rendering them using the code we managed to modify during the 3D programming. This had to be done using C++ to print out the data from our equations into txt files, as well as render it in the correct graph equations.

2 Tasks

2.1 Task 1

Following equation has been used:

$$f(x) = 2x^4 + 5x^3 - 0.4x - 0.6 \quad D_f = [-5, 5] \quad (1)$$

It's derivative is as follows:

$$f'(x) = 2x^3 + 5x^2 - 0.4 \quad (2)$$

To smoothen out the graph, $n = 100$ has been chosen. This made the resolution equal to:

$$h = \Delta x = \frac{5 - (-5)}{100} \quad (3)$$

Calculating x values for the polynomial and the derivative gives us these coordinates with the current resolution: (See Figure 1)

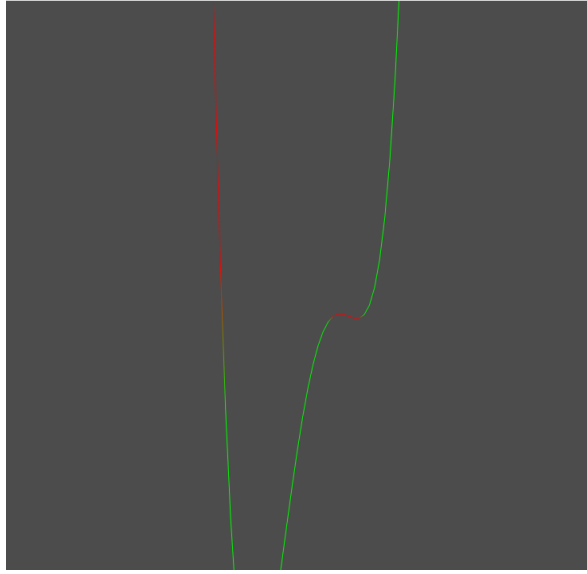


Figure 1: Generated Graph

```
#101
-5.000000 626.400024 0.000000 1.000000 0.000000 0.000000
-4.900000 566.075256 0.000000 1.000000 0.000000 0.000000
-4.800000 510.043304 0.000000 1.000000 0.000000 0.000000
-4.700000 458.101349 0.000000 1.000000 0.000000 0.000000
-4.600000 410.051361 0.000000 1.000000 0.000000 0.000000
-4.500000 365.700195 0.000000 1.000000 0.000000 0.000000
-4.400001 324.859436 0.000000 1.000000 0.000000 0.000000
-4.300001 287.345428 0.000000 1.000000 0.000000 0.000000
-4.200001 252.979446 0.000000 1.000000 0.000000 0.000000
-4.100001 221.587463 0.000000 1.000000 0.000000 0.000000
-4.000001 193.000259 0.000000 1.000000 0.000000 0.000000
-3.900001 167.053452 0.000000 1.000000 0.000000 0.000000
-3.800001 143.587448 0.000000 1.000000 0.000000 0.000000
-3.700001 122.447449 0.000000 1.000000 0.000000 0.000000
-3.600001 103.483437 0.000000 1.000000 0.000000 0.000000
```

Figure 2: First 15 lines of output in the generated text file

Listed below is the C++ code used to generate the vertices being rendered in the screenshot.

```
#include "oppg1.h"
#include <stdlib.h>

float f(const double x)
{
    return(x*x*x*x*x*2)+(5*x*x*x*x)-(0.4*x)-0.6;
}

float derf(const double x)
{
    return(2*x*x*x*x)+(5*x*x*x)-0.4;
}

std::vector<Vertex> getvertexes1()
{
    std::vector<Vertex> vertexes;
    DF df{-5, 5};
    unsigned int n = 100;
    float h = (df.b - df.a)/n;
    float x = df.a;
    float green, red;
```

```

    for (unsigned int i = 0; i < n + 1; i++){
        if (derf(x) > 0.0){
            green = 1.0;
            red = 0.0;
        }
        else {
            green = 0.0;
            red = 1.0;
        }
        vertexes.push_back(Vertex{x, f(x), 0.0f, red, green, 0.0f});
        x += h;
    }

    FILE* dataFile = fopen("Oppg1-Formula.txt", "w+");
    fprintf(dataFile, "%s%zu%s", "#", vertexes.size(), "\n");
    for (const Vertex& v : vertexes)
    {
        fprintf(dataFile, "%f-%f-%f-%f-%f-%f\n", v.x, v.y, v.z, v.r, v.g, v.b);
    }

    fclose(dataFile);
    return vertexes;
};

```

Which outputs a text document:

```

#101
-5.000000 626.400024 0.000000 1.000000 0.000000 0.000000
-4.900000 566.075256 0.000000 1.000000 0.000000 0.000000
-4.800000 510.043304 0.000000 1.000000 0.000000 0.000000
-4.700000 458.101349 0.000000 1.000000 0.000000 0.000000
-4.600000 410.051361 0.000000 1.000000 0.000000 0.000000
-4.500000 365.700195 0.000000 1.000000 0.000000 0.000000
-4.400001 324.859436 0.000000 1.000000 0.000000 0.000000
-4.300001 287.345428 0.000000 1.000000 0.000000 0.000000
-4.200001 252.979446 0.000000 1.000000 0.000000 0.000000
-4.100001 221.587463 0.000000 1.000000 0.000000 0.000000
-4.000001 193.000259 0.000000 1.000000 0.000000 0.000000
-3.900001 167.053452 0.000000 1.000000 0.000000 0.000000
-3.800001 143.587448 0.000000 1.000000 0.000000 0.000000
-3.700001 122.447449 0.000000 1.000000 0.000000 0.000000
-3.600001 103.483437 0.000000 1.000000 0.000000 0.000000
-3.500001 86.550232 0.000000 1.000000 0.000000 0.000000
-3.400002 71.507416 0.000000 1.000000 0.000000 0.000000
-3.300002 58.219402 0.000000 1.000000 0.000000 0.000000
-3.200002 46.555386 0.000000 1.000000 0.000000 0.000000
-3.100002 36.389370 0.000000 1.000000 0.000000 0.000000
-3.000002 27.600155 0.000000 1.000000 0.000000 0.000000
-2.900002 20.071339 0.000000 1.000000 0.000000 0.000000
-2.800002 13.691322 0.000000 1.000000 0.000000 0.000000
-2.700002 8.353307 0.000000 1.000000 0.000000 0.000000
-2.600002 3.955291 0.000000 1.000000 0.000000 0.000000
-2.500002 0.400075 0.000000 1.000000 0.000000 0.000000
-2.400002 -2.404739 0.000000 0.000000 1.000000 0.000000
-2.300003 -4.546752 0.000000 0.000000 1.000000 0.000000
-2.200003 -6.108765 0.000000 0.000000 1.000000 0.000000
-2.100003 -7.168777 0.000000 0.000000 1.000000 0.000000
-2.000003 -7.799987 0.000000 0.000000 1.000000 0.000000
-1.900003 -8.070797 0.000000 0.000000 1.000000 0.000000
-1.800003 -8.044805 0.000000 0.000000 1.000000 0.000000
-1.700003 -7.780810 0.000000 0.000000 1.000000 0.000000
-1.600003 -7.332815 0.000000 0.000000 1.000000 0.000000
-1.500003 -6.750018 0.000000 0.000000 1.000000 0.000000
-1.400003 -6.076819 0.000000 0.000000 1.000000 0.000000
-1.300003 -5.352820 0.000000 0.000000 1.000000 0.000000
-1.200003 -4.612820 0.000000 0.000000 1.000000 0.000000
-1.100003 -3.886819 0.000000 0.000000 1.000000 0.000000
-1.000003 -3.200017 0.000000 0.000000 1.000000 0.000000
-0.900003 -2.572815 0.000000 0.000000 1.000000 0.000000
-0.800003 -2.020813 0.000000 0.000000 1.000000 0.000000
-0.700003 -1.554811 0.000000 0.000000 1.000000 0.000000
-0.600003 -1.180808 0.000000 0.000000 1.000000 0.000000

```

-0.500003	-0.900006	0.000000	0.000000	1.000000	0.000000
-0.400003	-0.708804	0.000000	0.000000	1.000000	0.000000
-0.300003	-0.598802	0.000000	1.000000	0.000000	0.000000
-0.200003	-0.556800	0.000000	1.000000	0.000000	0.000000
-0.100003	-0.564799	0.000000	1.000000	0.000000	0.000000
-0.000003	-0.599999	0.000000	1.000000	0.000000	0.000000
0.099997	-0.634799	0.000000	1.000000	0.000000	0.000000
0.199997	-0.636801	0.000000	1.000000	0.000000	0.000000
0.299997	-0.568803	0.000000	0.000000	1.000000	0.000000
0.399997	-0.388806	0.000000	0.000000	1.000000	0.000000
0.499997	-0.050011	0.000000	0.000000	1.000000	0.000000
0.599997	0.499183	0.000000	0.000000	1.000000	0.000000
0.699997	1.315176	0.000000	0.000000	1.000000	0.000000
0.799998	2.459167	0.000000	0.000000	1.000000	0.000000
0.899998	3.997157	0.000000	0.000000	1.000000	0.000000
0.999998	5.999945	0.000000	0.000000	1.000000	0.000000
1.099998	8.543130	0.000000	0.000000	1.000000	0.000000
1.199998	11.707114	0.000000	0.000000	1.000000	0.000000
1.299998	15.577097	0.000000	0.000000	1.000000	0.000000
1.399998	20.243078	0.000000	0.000000	1.000000	0.000000
1.499998	25.799856	0.000000	0.000000	1.000000	0.000000
1.599998	32.347034	0.000000	0.000000	1.000000	0.000000
1.699998	39.989006	0.000000	0.000000	1.000000	0.000000
1.799998	48.834980	0.000000	0.000000	1.000000	0.000000
1.899998	58.998951	0.000000	0.000000	1.000000	0.000000
1.999998	70.599724	0.000000	0.000000	1.000000	0.000000
2.099998	83.760887	0.000000	0.000000	1.000000	0.000000
2.199998	98.610832	0.000000	0.000000	1.000000	0.000000
2.299998	115.282768	0.000000	0.000000	1.000000	0.000000
2.399997	133.914703	0.000000	0.000000	1.000000	0.000000
2.499997	154.649429	0.000000	0.000000	1.000000	0.000000
2.599997	177.634537	0.000000	0.000000	1.000000	0.000000
2.699997	203.022446	0.000000	0.000000	1.000000	0.000000
2.799997	230.970352	0.000000	0.000000	1.000000	0.000000
2.899997	261.640228	0.000000	0.000000	1.000000	0.000000
2.999997	295.198914	0.000000	0.000000	1.000000	0.000000
3.099997	331.817993	0.000000	0.000000	1.000000	0.000000
3.199997	371.673828	0.000000	0.000000	1.000000	0.000000
3.299997	414.947662	0.000000	0.000000	1.000000	0.000000
3.399997	461.825500	0.000000	0.000000	1.000000	0.000000
3.499996	512.498108	0.000000	0.000000	1.000000	0.000000
3.599996	567.161133	0.000000	0.000000	1.000000	0.000000
3.699996	626.014893	0.000000	0.000000	1.000000	0.000000
3.799996	689.264648	0.000000	0.000000	1.000000	0.000000
3.899996	757.120422	0.000000	0.000000	1.000000	0.000000
3.999996	829.796936	0.000000	0.000000	1.000000	0.000000
4.099996	907.514038	0.000000	0.000000	1.000000	0.000000
4.199996	990.495789	0.000000	0.000000	1.000000	0.000000
4.299996	1078.971436	0.000000	0.000000	1.000000	0.000000
4.399996	1173.175171	0.000000	0.000000	1.000000	0.000000
4.499996	1273.345581	0.000000	0.000000	1.000000	0.000000
4.599996	1379.726440	0.000000	0.000000	1.000000	0.000000
4.699996	1492.566040	0.000000	0.000000	1.000000	0.000000
4.799995	1612.117554	0.000000	0.000000	1.000000	0.000000
4.899995	1738.639160	0.000000	0.000000	1.000000	0.000000
4.999995	1872.393433	0.000000	0.000000	1.000000	0.000000

2.2 Task 2

We utilise the formula for a spiral:

$$f(x, y, z) = (\sin(t), x = \cos(t), \frac{t}{200}) \quad D_f = [0, 40] \quad (4)$$

We then use C++ to plot all vertices into a text document:

```
#include <stdlib.h>
#include "oppg2.h"

std::vector<Vertex> getvertexes2()
{
    float x = 0, y = 0, z = 0;
    std::vector<Vertex> vertexes;

    for (float i = 0; i < 10*4; i+=0.1)
    {
        x = sin(i);
        y = cos(i);
        z += 0.05;

        vertexes.push_back({x, y, z, 0.5, 0.5, 0.0});
    }

    FILE* dataFile = fopen("Oppg2-Formula.txt", "w+");
    fprintf(dataFile, "%s%zu%s", "#", vertexes.size(), "\n");
    for(const Vertex& v : vertexes)
    {
        fprintf(dataFile, "%f-%f-%f-%f-%f-%f\n", v.x, v.y, v.z, v.r, v.g, v.b);
    }

    fclose(dataFile);

    return vertexes;
}
}
```

Which outputs a text document:

```
#401
0.000000 1.000000 0.050000 0.500000 0.500000 0.000000
0.099833 0.995004 0.100000 0.500000 0.500000 0.000000
0.198669 0.980067 0.150000 0.500000 0.500000 0.000000
0.295520 0.955337 0.200000 0.500000 0.500000 0.000000
0.389418 0.921061 0.250000 0.500000 0.500000 0.000000
0.479426 0.877583 0.300000 0.500000 0.500000 0.000000
0.564642 0.825336 0.350000 0.500000 0.500000 0.000000
0.644218 0.764842 0.400000 0.500000 0.500000 0.000000
0.717356 0.696707 0.450000 0.500000 0.500000 0.000000
0.783327 0.621610 0.500000 0.500000 0.500000 0.000000
0.841471 0.540302 0.550000 0.500000 0.500000 0.000000
0.891207 0.453596 0.600000 0.500000 0.500000 0.000000
0.932039 0.362358 0.650000 0.500000 0.500000 0.000000
0.963558 0.267499 0.700000 0.500000 0.500000 0.000000
0.985450 0.169967 0.750000 0.500000 0.500000 0.000000
0.997495 0.070737 0.800000 0.500000 0.500000 0.000000
0.999574 -0.029200 0.850000 0.500000 0.500000 0.000000
0.991665 -0.128845 0.900000 0.500000 0.500000 0.000000
0.973848 -0.227202 0.950000 0.500000 0.500000 0.000000
0.946300 -0.323290 1.000000 0.500000 0.500000 0.000000
0.909297 -0.416147 1.050000 0.500000 0.500000 0.000000
0.863209 -0.504846 1.100000 0.500000 0.500000 0.000000
0.808496 -0.588501 1.150000 0.500000 0.500000 0.000000
0.745705 -0.666276 1.200000 0.500000 0.500000 0.000000
```

0.675463	-0.737394	1.250000	0.500000	0.500000	0.000000
0.598472	-0.801143	1.300000	0.500000	0.500000	0.000000
0.515502	-0.856889	1.350000	0.500000	0.500000	0.000000
0.427380	-0.904072	1.400000	0.500000	0.500000	0.000000
0.334989	-0.942222	1.450000	0.500000	0.500000	0.000000
0.239250	-0.970958	1.500000	0.500000	0.500000	0.000000
0.141121	-0.989992	1.550000	0.500000	0.500000	0.000000
0.041581	-0.999135	1.600000	0.500000	0.500000	0.000000
-0.058373	-0.998295	1.649999	0.500000	0.500000	0.000000
-0.157745	-0.987480	1.699999	0.500000	0.500000	0.000000
-0.255540	-0.966798	1.749999	0.500000	0.500000	0.000000
-0.350782	-0.936457	1.799999	0.500000	0.500000	0.000000
-0.442519	-0.896759	1.849999	0.500000	0.500000	0.000000
-0.529835	-0.848101	1.899999	0.500000	0.500000	0.000000
-0.611857	-0.790969	1.949999	0.500000	0.500000	0.000000
-0.687765	-0.725933	1.999999	0.500000	0.500000	0.000000
-0.756801	-0.653645	2.049999	0.500000	0.500000	0.000000
-0.818276	-0.574825	2.099999	0.500000	0.500000	0.000000
-0.871575	-0.490262	2.149999	0.500000	0.500000	0.000000
-0.916165	-0.400801	2.199999	0.500000	0.500000	0.000000
-0.951602	-0.307335	2.249999	0.500000	0.500000	0.000000
-0.977530	-0.210798	2.299999	0.500000	0.500000	0.000000
-0.993691	-0.112155	2.349999	0.500000	0.500000	0.000000
-0.999923	-0.012391	2.399999	0.500000	0.500000	0.000000
-0.996165	0.087497	2.449999	0.500000	0.500000	0.000000
-0.982453	0.186510	2.499999	0.500000	0.500000	0.000000
-0.958925	0.283660	2.549999	0.500000	0.500000	0.000000
-0.925816	0.377975	2.599999	0.500000	0.500000	0.000000
-0.883456	0.468514	2.649999	0.500000	0.500000	0.000000
-0.832269	0.554372	2.699999	0.500000	0.500000	0.000000
-0.772766	0.634691	2.749999	0.500000	0.500000	0.000000
-0.705542	0.708668	2.799999	0.500000	0.500000	0.000000
-0.631269	0.775564	2.849998	0.500000	0.500000	0.000000
-0.550688	0.834711	2.899998	0.500000	0.500000	0.000000
-0.464605	0.885518	2.949998	0.500000	0.500000	0.000000
-0.373880	0.927477	2.999998	0.500000	0.500000	0.000000
-0.279419	0.960169	3.049998	0.500000	0.500000	0.000000
-0.182166	0.983268	3.099998	0.500000	0.500000	0.000000
-0.083093	0.996542	3.149998	0.500000	0.500000	0.000000
0.016810	0.999859	3.199998	0.500000	0.500000	0.000000
0.116546	0.993185	3.249998	0.500000	0.500000	0.000000
0.215116	0.976588	3.299998	0.500000	0.500000	0.000000
0.311538	0.950234	3.349998	0.500000	0.500000	0.000000
0.404846	0.914385	3.399998	0.500000	0.500000	0.000000
0.494110	0.869399	3.449998	0.500000	0.500000	0.000000
0.578436	0.815728	3.499998	0.500000	0.500000	0.000000
0.656983	0.753905	3.549998	0.500000	0.500000	0.000000
0.728966	0.684550	3.599998	0.500000	0.500000	0.000000
0.793665	0.608355	3.649998	0.500000	0.500000	0.000000
0.850434	0.526081	3.699998	0.500000	0.500000	0.000000
0.898706	0.438552	3.749998	0.500000	0.500000	0.000000
0.937998	0.346640	3.799998	0.500000	0.500000	0.000000
0.967918	0.251265	3.849998	0.500000	0.500000	0.000000
0.988167	0.153379	3.899997	0.500000	0.500000	0.000000
0.998543	0.053960	3.949997	0.500000	0.500000	0.000000
0.998942	-0.045997	3.999997	0.500000	0.500000	0.000000
0.989359	-0.145495	4.049997	0.500000	0.500000	0.000000
0.969891	-0.243539	4.099998	0.500000	0.500000	0.000000
0.940732	-0.339150	4.149998	0.500000	0.500000	0.000000
0.902174	-0.431373	4.199998	0.500000	0.500000	0.000000
0.854601	-0.519285	4.249998	0.500000	0.500000	0.000000
0.798489	-0.602009	4.299998	0.500000	0.500000	0.000000
0.734399	-0.678718	4.349998	0.500000	0.500000	0.000000
0.662971	-0.748645	4.399999	0.500000	0.500000	0.000000
0.584919	-0.811091	4.449999	0.500000	0.500000	0.000000
0.501023	-0.865434	4.499999	0.500000	0.500000	0.000000
0.412120	-0.911129	4.549999	0.500000	0.500000	0.000000
0.319100	-0.947721	4.599999	0.500000	0.500000	0.000000
0.222891	-0.974843	4.650000	0.500000	0.500000	0.000000
0.124455	-0.992225	4.700000	0.500000	0.500000	0.000000
0.024776	-0.999693	4.750000	0.500000	0.500000	0.000000

-0.075151	-0.997172	4.800000	0.500000	0.500000	0.000000
-0.174327	-0.984688	4.850000	0.500000	0.500000	0.000000
-0.271761	-0.962365	4.900001	0.500000	0.500000	0.000000
-0.366480	-0.930426	4.950001	0.500000	0.500000	0.000000
-0.457537	-0.889190	5.000001	0.500000	0.500000	0.000000
-0.544023	-0.839070	5.050001	0.500000	0.500000	0.000000
-0.625072	-0.780567	5.100001	0.500000	0.500000	0.000000
-0.699877	-0.714264	5.150002	0.500000	0.500000	0.000000
-0.767688	-0.640824	5.200002	0.500000	0.500000	0.000000
-0.827828	-0.560981	5.250002	0.500000	0.500000	0.000000
-0.879698	-0.475534	5.300002	0.500000	0.500000	0.000000
-0.922777	-0.385334	5.350002	0.500000	0.500000	0.000000
-0.956636	-0.291285	5.400002	0.500000	0.500000	0.000000
-0.980937	-0.194325	5.450003	0.500000	0.500000	0.000000
-0.995437	-0.095424	5.500003	0.500000	0.500000	0.000000
-0.999990	0.004431	5.550003	0.500000	0.500000	0.000000
-0.994552	0.104242	5.600003	0.500000	0.500000	0.000000
-0.979176	0.203011	5.650003	0.500000	0.500000	0.000000
-0.954017	0.299752	5.700004	0.500000	0.500000	0.000000
-0.919326	0.393498	5.750004	0.500000	0.500000	0.000000
-0.875448	0.483311	5.800004	0.500000	0.500000	0.000000
-0.822824	0.568296	5.850004	0.500000	0.500000	0.000000
-0.761978	0.647603	5.900004	0.500000	0.500000	0.000000
-0.693519	0.720439	5.950005	0.500000	0.500000	0.000000
-0.618130	0.786076	6.000005	0.500000	0.500000	0.000000
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-0.358220	0.933637	6.150005	0.500000	0.500000	0.000000
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0.231522	0.972830	6.450006	0.500000	0.500000	0.000000
0.327487	0.944856	6.500007	0.500000	0.500000	0.000000
0.420179	0.907441	6.550007	0.500000	0.500000	0.000000
0.508673	0.860960	6.600007	0.500000	0.500000	0.000000
0.592085	0.805876	6.650007	0.500000	0.500000	0.000000
0.669581	0.742739	6.700007	0.500000	0.500000	0.000000
0.740386	0.672182	6.750008	0.500000	0.500000	0.000000
0.803793	0.594908	6.800008	0.500000	0.500000	0.000000
0.859170	0.511691	6.850008	0.500000	0.500000	0.000000
0.905962	0.423360	6.900008	0.500000	0.500000	0.000000
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0.998026	-0.062810	7.150009	0.500000	0.500000	0.000000
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-0.840984	0.541060	18.400005	0.500000	0.500000	0.000000
-0.782768	0.622314	18.450005	0.500000	0.500000	0.000000
-0.716731	0.697350	18.500004	0.500000	0.500000	0.000000
-0.643532	0.765419	18.550003	0.500000	0.500000	0.000000
-0.563904	0.825840	18.600002	0.500000	0.500000	0.000000
-0.478642	0.878010	18.650002	0.500000	0.500000	0.000000
-0.388597	0.921408	18.700001	0.500000	0.500000	0.000000
-0.294670	0.955599	18.750000	0.500000	0.500000	0.000000
-0.197799	0.980243	18.799999	0.500000	0.500000	0.000000
-0.098951	0.995092	18.849998	0.500000	0.500000	0.000000
0.000885	1.000000	18.899998	0.500000	0.500000	0.000000
0.100713	0.994916	18.949997	0.500000	0.500000	0.000000

0.199534	0.979891	18.999996	0.500000	0.500000	0.000000
0.296361	0.955076	19.049995	0.500000	0.500000	0.000000
0.390228	0.920718	19.099995	0.500000	0.500000	0.000000
0.480195	0.877162	19.149994	0.500000	0.500000	0.000000
0.565365	0.824841	19.199993	0.500000	0.500000	0.000000
0.644886	0.764279	19.249992	0.500000	0.500000	0.000000
0.717964	0.696080	19.299992	0.500000	0.500000	0.000000
0.783868	0.620927	19.349991	0.500000	0.500000	0.000000
0.841941	0.539570	19.399990	0.500000	0.500000	0.000000
0.891601	0.452822	19.449989	0.500000	0.500000	0.000000
0.932353	0.361550	19.499989	0.500000	0.500000	0.000000
0.963789	0.266665	19.549988	0.500000	0.500000	0.000000
0.985596	0.169116	19.599987	0.500000	0.500000	0.000000
0.997556	0.069877	19.649986	0.500000	0.500000	0.000000
0.999548	-0.030060	19.699986	0.500000	0.500000	0.000000
0.991554	-0.129696	19.749985	0.500000	0.500000	0.000000
0.973652	-0.228037	19.799984	0.500000	0.500000	0.000000
0.946023	-0.324100	19.849983	0.500000	0.500000	0.000000
0.908941	-0.416924	19.899982	0.500000	0.500000	0.000000
0.862778	-0.505582	19.949982	0.500000	0.500000	0.000000
0.807995	-0.589189	19.999981	0.500000	0.500000	0.000000
0.745139	-0.666910	20.049980	0.500000	0.500000	0.000000

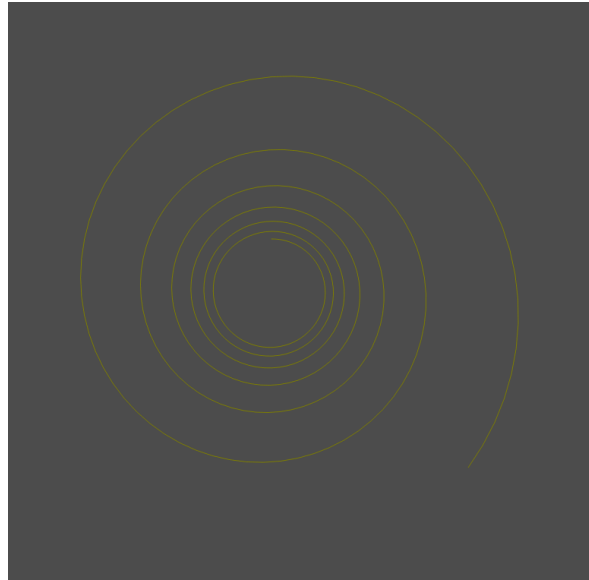


Figure 3: Spiral rendered from the vertex output from our C++ code

2.3 Task 3

In this task we use a function called apesadelen:

$$f(x) = x^2y \quad (5)$$

For this task I have used the same parameters as the example in section 2.7 in the lecture notes, which when put into C++ becomes this:

```
#include "oppg3.h"
#include <cmath.h>

double f(const float x, const float y){
    return x*x*y;
};

std::vector<Vertex> getvertexes3()
{
    std::vector<Vertex> vertexes;

    float xmin=0.0f, xmax=1.0f, ymin=0.0f, ymax=1.0f, h=0.125f;
    for (auto x=xmin; x<xmax; x+=h) {
        for (auto y=ymin; y<ymax; y+=h)
        {
            float z = f(x, y);
            vertexes.push_back(Vertex{x,y,z,x,y,z});
            z = f(x+h, y);
            vertexes.push_back(Vertex{x+h,y,z,x,y,z});
            z = f(x, y+h);
            vertexes.push_back(Vertex{x,y+h,z,x,y,z});
            vertexes.push_back(Vertex{x,y+h,z,x,y,z});
            z = f(x+h, y);
            vertexes.push_back(Vertex{x+h,y,z,x,y,z});
            z = f(x+h, y+h);
            vertexes.push_back(Vertex{x+h,y+h,z,x,y,z});
        }
    }

    FILE* dataFile = fopen("Oppg3_Formula.txt", "w+");
    fprintf(dataFile, "%s%zu%s", "#", vertexes.size(), "\n");
    for(const Vertex& v : vertexes)
```

```

    {
        fprintf(dataFile, "%f-%f-%f-%f-%f-%f\n", v.x, v.y, v.z, v.r, v.g, v.b);
    }

    fclose(dataFile);

    return vertices;
}

```

Which outputs a text document:

```

#384
0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
0.125000 0.000000 0.000000 0.000000 0.000000 0.000000
0.000000 0.125000 0.000000 0.000000 0.000000 0.000000
0.000000 0.125000 0.000000 0.000000 0.000000 0.000000
0.125000 0.000000 0.000000 0.000000 0.000000 0.000000
0.125000 0.125000 0.001953 0.000000 0.000000 0.001953
0.000000 0.125000 0.000000 0.000000 0.125000 0.000000
0.125000 0.125000 0.001953 0.000000 0.125000 0.001953
0.000000 0.250000 0.000000 0.000000 0.125000 0.000000
0.000000 0.250000 0.000000 0.000000 0.125000 0.000000
0.125000 0.125000 0.001953 0.000000 0.125000 0.001953
0.125000 0.250000 0.003906 0.000000 0.125000 0.003906
0.000000 0.250000 0.000000 0.000000 0.250000 0.000000
0.000000 0.250000 0.000000 0.000000 0.250000 0.000000
0.125000 0.250000 0.003906 0.000000 0.250000 0.003906
0.000000 0.375000 0.000000 0.000000 0.250000 0.000000
0.000000 0.375000 0.000000 0.000000 0.250000 0.000000
0.125000 0.250000 0.003906 0.000000 0.250000 0.003906
0.125000 0.375000 0.005859 0.000000 0.250000 0.005859
0.000000 0.375000 0.000000 0.000000 0.375000 0.000000
0.125000 0.375000 0.005859 0.000000 0.375000 0.005859
0.000000 0.500000 0.000000 0.000000 0.375000 0.000000
0.000000 0.500000 0.000000 0.000000 0.375000 0.000000
0.125000 0.375000 0.005859 0.000000 0.375000 0.005859
0.125000 0.500000 0.007812 0.000000 0.375000 0.007812
0.000000 0.500000 0.000000 0.000000 0.500000 0.000000
0.125000 0.500000 0.007812 0.000000 0.500000 0.007812
0.000000 0.625000 0.000000 0.000000 0.500000 0.000000
0.000000 0.625000 0.000000 0.000000 0.500000 0.000000
0.125000 0.500000 0.007812 0.000000 0.500000 0.007812
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0.000000 0.750000 0.000000 0.000000 0.625000 0.000000
0.000000 0.750000 0.000000 0.000000 0.625000 0.000000
0.125000 0.625000 0.009766 0.000000 0.625000 0.009766
0.125000 0.750000 0.011719 0.000000 0.625000 0.011719
0.000000 0.750000 0.000000 0.000000 0.750000 0.000000
0.125000 0.750000 0.011719 0.000000 0.750000 0.011719
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0.000000 0.875000 0.000000 0.000000 0.750000 0.000000
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0.125000 0.000000 0.000000 0.125000 0.000000 0.000000
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0.125000 0.125000 0.001953 0.125000 0.125000 0.001953
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0.125000 0.250000 0.003906 0.125000 0.125000 0.003906
0.125000 0.250000 0.003906 0.125000 0.125000 0.003906
0.250000 0.125000 0.007812 0.125000 0.125000 0.007812

```

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0.125000	0.375000	0.005859	0.125000	0.250000	0.005859
0.125000	0.375000	0.005859	0.125000	0.250000	0.005859
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0.250000	0.375000	0.023438	0.125000	0.250000	0.023438
0.125000	0.375000	0.005859	0.125000	0.375000	0.005859
0.250000	0.375000	0.023438	0.125000	0.375000	0.023438
0.125000	0.500000	0.007812	0.125000	0.375000	0.007812
0.125000	0.500000	0.007812	0.125000	0.375000	0.007812
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0.250000	0.125000	0.007812	0.250000	0.000000	0.007812
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0.250000	0.125000	0.007812	0.250000	0.125000	0.007812
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0.250000	0.250000	0.015625	0.250000	0.125000	0.015625
0.250000	0.250000	0.015625	0.250000	0.125000	0.015625
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0.375000	0.250000	0.035156	0.250000	0.250000	0.035156
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0.375000	0.250000	0.035156	0.250000	0.250000	0.035156
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0.250000	0.375000	0.023438	0.250000	0.375000	0.023438
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0.250000	0.500000	0.031250	0.250000	0.375000	0.031250
0.375000	0.375000	0.052734	0.250000	0.375000	0.052734
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0.250000	0.750000	0.046875	0.250000	0.625000	0.046875

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0.375000	0.750000	0.105469	0.250000	0.750000	0.105469
0.250000	0.875000	0.054688	0.250000	0.750000	0.054688
0.250000	0.875000	0.054688	0.250000	0.750000	0.054688
0.375000	0.750000	0.105469	0.250000	0.750000	0.105469
0.375000	0.875000	0.123047	0.250000	0.750000	0.123047
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0.250000	1.000000	0.062500	0.250000	0.875000	0.062500
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0.500000	0.000000	0.000000	0.375000	0.000000	0.000000
0.500000	0.125000	0.031250	0.375000	0.000000	0.031250
0.375000	0.125000	0.017578	0.375000	0.125000	0.017578
0.500000	0.125000	0.031250	0.375000	0.125000	0.031250
0.375000	0.250000	0.035156	0.375000	0.125000	0.035156
0.375000	0.250000	0.035156	0.375000	0.125000	0.035156
0.500000	0.125000	0.031250	0.375000	0.125000	0.031250
0.500000	0.250000	0.062500	0.375000	0.125000	0.062500
0.375000	0.250000	0.035156	0.375000	0.250000	0.035156
0.500000	0.250000	0.062500	0.375000	0.250000	0.062500
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0.375000	0.375000	0.052734	0.375000	0.250000	0.052734
0.500000	0.250000	0.062500	0.375000	0.250000	0.062500
0.500000	0.375000	0.093750	0.375000	0.250000	0.093750
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0.375000	0.625000	0.087891	0.375000	0.500000	0.087891
0.375000	0.625000	0.087891	0.375000	0.500000	0.087891
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0.500000	0.625000	0.156250	0.375000	0.625000	0.156250
0.375000	0.750000	0.105469	0.375000	0.625000	0.105469
0.375000	0.750000	0.105469	0.375000	0.625000	0.105469
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0.500000	0.750000	0.187500	0.375000	0.750000	0.187500
0.375000	0.875000	0.123047	0.375000	0.750000	0.123047
0.375000	0.875000	0.123047	0.375000	0.750000	0.123047
0.500000	0.750000	0.187500	0.375000	0.750000	0.187500
0.500000	0.875000	0.218750	0.375000	0.750000	0.218750
0.375000	0.875000	0.123047	0.375000	0.875000	0.123047
0.500000	0.875000	0.218750	0.375000	0.875000	0.218750
0.375000	1.000000	0.140625	0.375000	0.875000	0.140625
0.375000	1.000000	0.140625	0.375000	0.875000	0.140625
0.500000	0.875000	0.218750	0.375000	0.875000	0.218750
0.500000	1.000000	0.250000	0.375000	0.875000	0.250000
0.500000	0.000000	0.000000	0.500000	0.000000	0.000000
0.625000	0.000000	0.000000	0.500000	0.000000	0.000000
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0.625000	0.125000	0.048828	0.500000	0.125000	0.048828
0.500000	0.250000	0.062500	0.500000	0.125000	0.062500

0.500000	0.250000	0.062500	0.500000	0.125000	0.062500
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0.625000	0.250000	0.097656	0.500000	0.250000	0.097656
0.500000	0.375000	0.093750	0.500000	0.250000	0.093750
0.500000	0.375000	0.093750	0.500000	0.250000	0.093750
0.625000	0.250000	0.097656	0.500000	0.250000	0.097656
0.625000	0.375000	0.146484	0.500000	0.250000	0.146484
0.500000	0.375000	0.093750	0.500000	0.375000	0.093750
0.625000	0.375000	0.146484	0.500000	0.375000	0.146484
0.500000	0.500000	0.125000	0.500000	0.375000	0.125000
0.500000	0.500000	0.125000	0.500000	0.375000	0.125000
0.625000	0.375000	0.146484	0.500000	0.375000	0.146484
0.625000	0.500000	0.195312	0.500000	0.375000	0.195312
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0.500000	0.625000	0.156250	0.500000	0.500000	0.156250
0.625000	0.500000	0.195312	0.500000	0.500000	0.195312
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0.500000	0.625000	0.156250	0.500000	0.625000	0.156250
0.625000	0.625000	0.244141	0.500000	0.625000	0.244141
0.500000	0.750000	0.187500	0.500000	0.625000	0.187500
0.500000	0.750000	0.187500	0.500000	0.625000	0.187500
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0.625000	0.750000	0.292969	0.500000	0.625000	0.292969
0.500000	0.750000	0.187500	0.500000	0.750000	0.187500
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0.500000	0.875000	0.218750	0.500000	0.750000	0.218750
0.500000	0.875000	0.218750	0.500000	0.750000	0.218750
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0.625000	0.875000	0.341797	0.500000	0.750000	0.341797
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0.750000	0.000000	0.000000	0.625000	0.000000	0.000000
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0.750000	0.125000	0.070312	0.625000	0.125000	0.070312
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0.625000	0.250000	0.097656	0.625000	0.125000	0.097656
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0.625000	0.375000	0.146484	0.625000	0.250000	0.146484
0.625000	0.375000	0.146484	0.625000	0.250000	0.146484
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0.625000	0.625000	0.244141	0.625000	0.625000	0.244141
0.750000	0.625000	0.351562	0.625000	0.625000	0.351562

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0.625000	0.750000	0.292969	0.625000	0.625000	0.292969
0.750000	0.625000	0.351562	0.625000	0.625000	0.351562
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0.625000	0.750000	0.292969	0.625000	0.750000	0.292969
0.750000	0.750000	0.421875	0.625000	0.750000	0.421875
0.625000	0.875000	0.341797	0.625000	0.750000	0.341797
0.625000	0.875000	0.341797	0.625000	0.750000	0.341797
0.750000	0.750000	0.421875	0.625000	0.750000	0.421875
0.750000	0.875000	0.492188	0.625000	0.750000	0.492188
0.625000	0.875000	0.341797	0.625000	0.875000	0.341797
0.750000	0.875000	0.492188	0.625000	0.875000	0.492188
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0.625000	1.000000	0.390625	0.625000	0.875000	0.390625
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0.750000	0.125000	0.070312	0.750000	0.000000	0.070312
0.875000	0.000000	0.000000	0.750000	0.000000	0.000000
0.875000	0.125000	0.095703	0.750000	0.000000	0.095703
0.750000	0.125000	0.070312	0.750000	0.125000	0.070312
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0.750000	0.250000	0.140625	0.750000	0.125000	0.140625
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0.750000	0.375000	0.210938	0.750000	0.250000	0.210938
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0.875000	0.375000	0.287109	0.750000	0.250000	0.287109
0.750000	0.375000	0.210938	0.750000	0.375000	0.210938
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0.750000	0.500000	0.281250	0.750000	0.375000	0.281250
0.750000	0.500000	0.281250	0.750000	0.375000	0.281250
0.875000	0.375000	0.287109	0.750000	0.375000	0.287109
0.875000	0.500000	0.382812	0.750000	0.375000	0.382812
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0.750000	0.625000	0.351562	0.750000	0.500000	0.351562
0.750000	0.625000	0.351562	0.750000	0.500000	0.351562
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0.875000	0.625000	0.478516	0.750000	0.625000	0.478516
0.750000	0.750000	0.421875	0.750000	0.625000	0.421875
0.750000	0.750000	0.421875	0.750000	0.625000	0.421875
0.875000	0.625000	0.478516	0.750000	0.625000	0.478516
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0.750000	0.750000	0.421875	0.750000	0.750000	0.421875
0.875000	0.750000	0.574219	0.750000	0.750000	0.574219
0.750000	0.875000	0.492188	0.750000	0.750000	0.492188
0.750000	0.875000	0.492188	0.750000	0.750000	0.492188
0.875000	0.750000	0.574219	0.750000	0.750000	0.574219
0.875000	0.875000	0.669922	0.750000	0.750000	0.669922
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0.750000	1.000000	0.562500	0.750000	0.875000	0.562500
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1.000000	0.000000	0.000000	0.875000	0.000000	0.000000
1.000000	0.125000	0.095703	0.875000	0.000000	0.125000
0.875000	0.125000	0.095703	0.875000	0.125000	0.095703

1.000000	0.125000	0.125000	0.875000	0.125000	0.125000
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0.875000	0.250000	0.191406	0.875000	0.125000	0.191406
1.000000	0.125000	0.125000	0.875000	0.125000	0.125000
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0.875000	0.250000	0.191406	0.875000	0.250000	0.191406
1.000000	0.250000	0.250000	0.875000	0.250000	0.250000
0.875000	0.375000	0.287109	0.875000	0.250000	0.287109
0.875000	0.375000	0.287109	0.875000	0.250000	0.287109
1.000000	0.250000	0.250000	0.875000	0.250000	0.250000
1.000000	0.375000	0.375000	0.875000	0.250000	0.375000
0.875000	0.375000	0.287109	0.875000	0.375000	0.287109
1.000000	0.375000	0.375000	0.875000	0.375000	0.375000
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0.875000	0.500000	0.382812	0.875000	0.375000	0.382812
1.000000	0.375000	0.375000	0.875000	0.375000	0.375000
1.000000	0.500000	0.500000	0.875000	0.375000	0.500000
0.875000	0.500000	0.382812	0.875000	0.500000	0.382812
1.000000	0.500000	0.500000	0.875000	0.500000	0.500000
0.875000	0.625000	0.478516	0.875000	0.500000	0.478516
0.875000	0.625000	0.478516	0.875000	0.500000	0.478516
1.000000	0.500000	0.500000	0.875000	0.500000	0.500000
1.000000	0.625000	0.625000	0.875000	0.500000	0.625000
0.875000	0.625000	0.478516	0.875000	0.625000	0.478516
1.000000	0.625000	0.625000	0.875000	0.625000	0.625000
0.875000	0.750000	0.574219	0.875000	0.625000	0.574219
0.875000	0.750000	0.574219	0.875000	0.625000	0.574219
1.000000	0.625000	0.625000	0.875000	0.625000	0.625000
1.000000	0.750000	0.750000	0.875000	0.625000	0.750000
0.875000	0.750000	0.574219	0.875000	0.750000	0.574219
1.000000	0.750000	0.750000	0.875000	0.750000	0.750000
0.875000	0.875000	0.669922	0.875000	0.750000	0.669922
0.875000	0.875000	0.669922	0.875000	0.750000	0.669922
1.000000	0.750000	0.750000	0.875000	0.750000	0.750000
1.000000	0.875000	0.875000	0.875000	0.750000	0.875000
0.875000	0.875000	0.669922	0.875000	0.875000	0.669922
1.000000	0.875000	0.875000	0.875000	0.875000	0.875000
0.875000	1.000000	0.765625	0.875000	0.875000	0.765625
0.875000	1.000000	0.765625	0.875000	0.875000	0.765625
1.000000	0.875000	0.875000	0.875000	0.875000	0.875000
1.000000	1.000000	1.000000	0.875000	0.875000	1.000000

With this text document, we can read it with our rendering software, giving us this rendered output: (See figure 4)

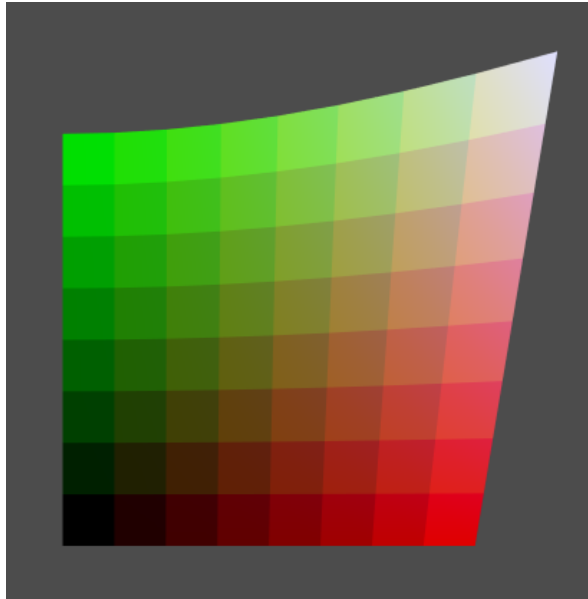


Figure 4: Shape rendered from vertex output of the C++ code

3 Discussion

I understood what we were supposed to do in the end with the little help of my classmates. On a side note, while troubleshooting my rendering issues, I ended up reading much more Vulkan forum discussions that I ever did on any other subject. That was... new.

I now know that vertices is not, in fact, written with an x.